

c) For some reasonable parameters and initial conditions, answer the question about braking stability.

Using parameters...

$$g = 10$$

$$I_G = 1$$

$$m = 1$$

$$d1 = -0.5$$

$$d2 = -0.5$$

$$\mu_k = 0.3$$

And initial conditions...

$$\omega = 0.5$$

$$v_{c0} = 10$$

and setting up such that the skate (free rolling wheels) is leading, we see that after some time the locked wheels will spin around and then lead the path of motion until the system comes to a complete stop (see figures 1 and 2).

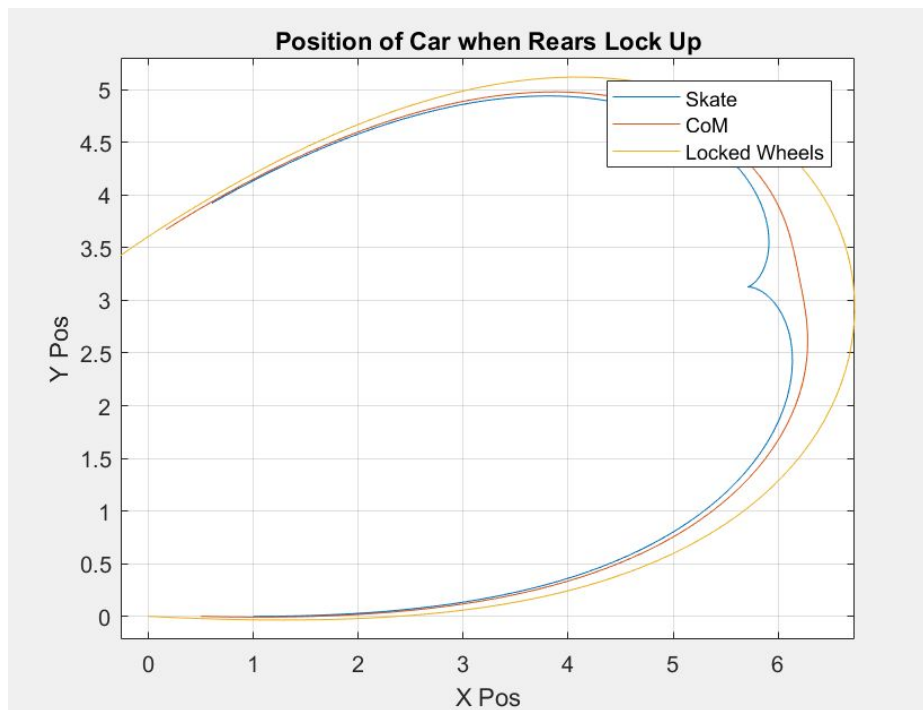


Figure 1

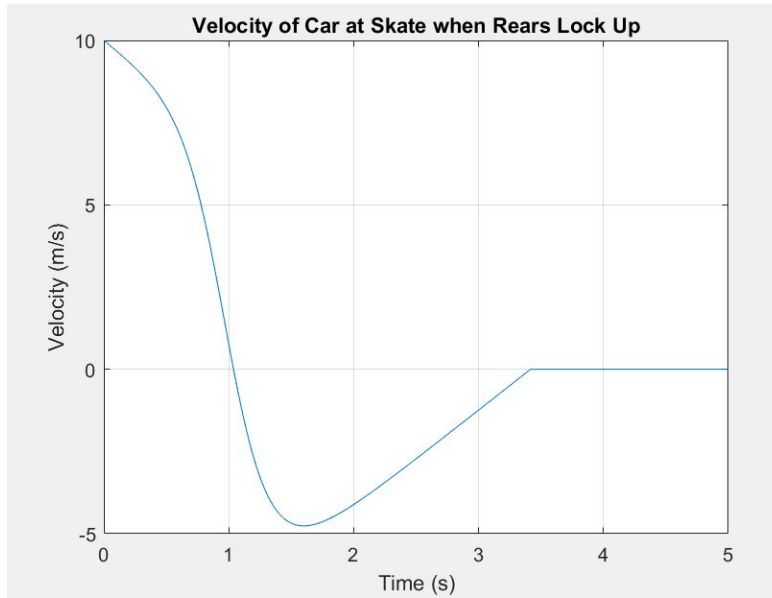


Figure 2

Now consider the same situation if the front brakes lock up; the skid point leads in front of the skate. The system will stably come to a complete stop, slightly veering of a straight path, which is an effect of the initial conditions (see figures 3 and 4).

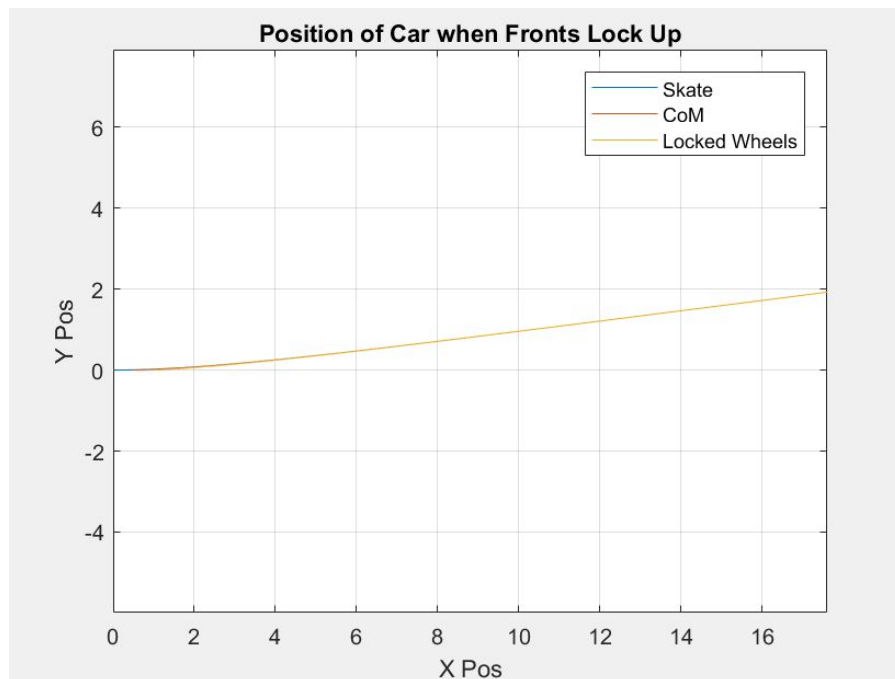


Figure 3

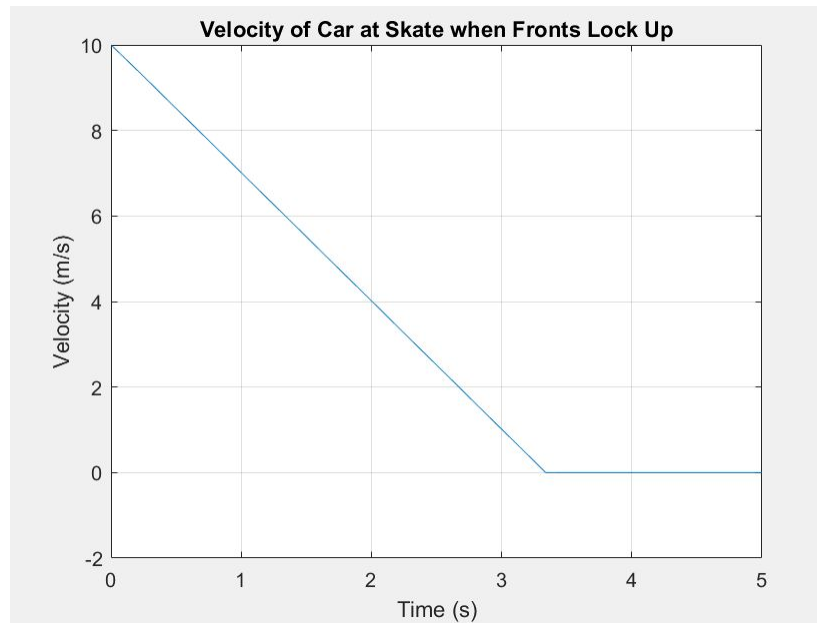


Figure 4

Based on these results, braking stability of a car is much better if the front wheels lock up and catastrophic if the rear wheels lock up, given the initial forward velocity and angular velocity of the car.